

## Literature-Already-Answered Packet

Upper Khintchine estimate in noncommutative symmetric spaces

**Status:** literature already answered, not a new automatic-discovery proof

### Source and later answer

The source paper is:

arXiv:0803.4404, Christian Le Merdy and Fedor Sukochev,

*Rademacher averages on noncommutative symmetric spaces*, 2008.

It appeared in *Journal of Functional Analysis* 255 (2008), 3329–3355. The local copy is included as `source_paper.pdf`.

The later supporting paper is:

S. Dirksen, B. de Pagter, D. Potapov, and F. Sukochev,

*Rosenthal inequalities in noncommutative symmetric spaces*,

*Journal of Functional Analysis* 261 (2011), no. 10, 2890–2925,

DOI: 10.1016/j.jfa.2011.07.015.

### Open question in the source

Let  $E$  be a separable, or dual-separable, symmetric function space, let  $M$  be a semifinite von Neumann algebra, and let  $E(M)$  be the associated noncommutative symmetric space. For a finite family  $(x_k) \subset E(M)$ , the source defines

$$\|(x_k)\|_{\max} = \max \left\{ \left\| \left( \sum_k x_k^* x_k \right)^{1/2} \right\|_{E(M)}, \left\| \left( \sum_k x_k x_k^* \right)^{1/2} \right\|_{E(M)} \right\}.$$

Theorem 1.1(2) of the source proves, under  $q_E < \infty$  and  $p_E > 1$ , the upper Rademacher estimate

$$\left\| \sum_k \varepsilon_k \otimes x_k \right\|_E \lesssim_E \|(x_k)\|_{\max}. \quad (1)$$

Immediately after Theorem 1.1, in the introduction, the authors state as an open problem that they do not know whether this second part remains true without assuming  $p_E > 1$ .

## Supporting answer

The abstract of Dirksen–de Pagter–Potapov–Sukochev (2011) states that the paper gives a direct proof of the upper Khintchine inequality for a noncommutative symmetric quasi-Banach function space with nontrivial upper Boyd index, and that this settles an open question of Le Merdy and Sukochev (2008).

The condition “nontrivial upper Boyd index” is the condition  $q_E < \infty$  in the notation of the source paper. Thus the supporting result proves the upper Khintchine estimate (1) under the source paper’s  $q_E < \infty$  hypothesis alone, removing the extra assumption  $p_E > 1$ .

**Claim.** *The open problem after Theorem 1.1 in arXiv:0803.4404 is already answered by Dirksen–de Pagter–Potapov–Sukochev, Rosenthal inequalities in noncommutative symmetric spaces, JFA 261 (2011), 2890–2925.*

*Proof.* This is a direct literature identification. The source question asks exactly whether the upper Khintchine/Rademacher estimate in Theorem 1.1(2) remains true when the assumption  $p_E > 1$  is removed while retaining  $q_E < \infty$ . The supporting paper explicitly says that it proves the upper Khintchine inequality for noncommutative symmetric quasi-Banach function spaces with nontrivial upper Boyd index and that this settles the Le Merdy–Sukochev open question. Since nontrivial upper Boyd index is  $q_E < \infty$ , this is the requested removal of  $p_E > 1$ . Therefore the extracted open problem has a separate later literature answer.  $\square$

## Classification and scope

This packet belongs in `solutions/literature.already_answered/`. The supporting authors explicitly identify their result as settling the open question from Le Merdy–Sukochev (2008). This is a retrieval/status result, not a new proof by the run.

The recorded answer is limited to the upper Khintchine estimate in Theorem 1.1(2). The source paper also discusses examples, optimality issues, and double sums; this packet makes no additional claim about those passages unless checked separately.

## Verification notes

- The source is not a same-paper false positive: the open problem appears immediately after Theorem 1.1 in the 2008 paper.
- The supporting source is separate and later, and its abstract explicitly says that it settles the Le Merdy–Sukochev open question.
- The decisive match is between  $q_E < \infty$  in the source theorem and nontrivial upper Boyd index in the supporting paper.

## References

- C. Le Merdy and F. Sukochev, arXiv:0803.4404, *Rademacher averages on noncommutative symmetric spaces*, J. Funct. Anal. 255 (2008), 3329–3355.
- S. Dirksen, B. de Pagter, D. Potapov, and F. Sukochev, *Rosenthal inequalities in noncommutative symmetric spaces*, J. Funct. Anal. 261 (2011), no. 10, 2890–2925.  
DOI: 10.1016/j.jfa.2011.07.015.